Advanced Econometrics I
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Problem Set 1 – Suggested Answers

1. **Cauchy sequence on** $(0, 1]$.
   Consider $Z = (0, 1]$ with $d(x, y) = |x - y|$ and $x_n = 1/n$.
   Is it a Cauchy sequence? Yes.

   Proof. Let $\varepsilon > 0$. Choose $N > 2/\varepsilon$. For all $m, n > N$:
   $$d(x_m, x_n) = \left|\frac{1}{m} - \frac{1}{n}\right| \leq \frac{1}{m} + \frac{1}{n} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$
   Hence (x_n) is Cauchy. □

   Does it converge in Z? No.

   Proof. Suppose $x_n \to x$ for some $x \in Z$. Then $d(x_n, x) = |1/n - x| \to 0$, which forces $x = 0$. But $0 \notin (0, 1]$, a contradiction. Hence the sequence does not converge in Z. □

   Discussion. This is the key example illustrating why *completeness* matters. The space $(0, 1]$ is not complete: there exists a Cauchy sequence with no limit in the space. The "missing" limit point is 0, which was excluded. In contrast, $[0, 1]$ with the same metric is complete. The takeaway: in incomplete spaces, sequences can "try to converge" to a point outside the space.

2. **Cauchy–Schwarz and Triangle inequality.**
   Let X, Y be elements of a Hilbert space.

   (i) **Cauchy–Schwarz:** $|\langle X, Y\rangle| \leq \|X\| \cdot \|Y\|$.

   Proof. If $Y = 0$, then both sides equal zero and the inequality holds trivially. Now assume $Y \neq 0$, so $\|Y\| > 0$. For any $t \in \mathbb{R}$, the non-negativity of the norm-squared gives:
   $$\begin{aligned} 0 \leq \|X - tY\|^2 &= \langle X - tY, X - tY\rangle \\ &= \|X\|^2 - 2t\langle X, Y\rangle + t^2 \|Y\|^2 . \end{aligned}$$
   This is a quadratic in t that is non-negative for all $t \in \mathbb{R}$. Choose $t^* = \langle X, Y\rangle / \|Y\|^2$ (the value that minimizes the quadratic—note this is exactly the projection coefficient of X onto Y). Substituting:
   $$0 \leq \|X\|^2 - 2\frac{\langle X, Y\rangle}{\|Y\|^2}\langle X, Y\rangle + \frac{\langle X, Y\rangle^2}{\|Y\|^4} \|Y\|^2 = \|X\|^2 - \frac{\langle X, Y\rangle^2}{\|Y\|^2}.$$
   Rearranging: $\langle X, Y\rangle^2 \leq \|X\|^2 \|Y\|^2$. Taking square roots (both sides non-negative):
   $$|\langle X, Y\rangle| \leq \|X\| \cdot \|Y\| \square$$

Remark. The optimal $t^* = \langle X, Y\rangle / \|Y\|^2$ is the projection coefficient of X onto Y. The proof shows that the "residual" $X - t^*Y$ has non-negative squared norm, which is equivalent to Cauchy–Schwarz. This connection between Cauchy–Schwarz and projection will recur throughout the course.

(ii) **Triangle inequality:** $\|X + Y\| \le \|X\| + \|Y\|$.

Proof. Expand the squared norm:

$$\|X + Y\|^2 = \langle X + Y, X + Y\rangle = \|X\|^2 + 2\langle X, Y\rangle + \|Y\|^2 .$$

By the Cauchy–Schwarz inequality: $\langle X, Y\rangle \le |\langle X, Y\rangle| \le \|X\| \, \|Y\|$. Hence:

$$\|X + Y\|^2 \le \|X\|^2 + 2 \, \|X\| \, \|Y\| + \|Y\|^2 = (\|X\| + \|Y\|)^2 .$$

Taking square roots (both sides non-negative): $\|X + Y\| \le \|X\| + \|Y\|$. □

3. **Finite second moments imply finite covariances.**

Assume $\mathrm{E}(X^2) < \infty$ and $\mathrm{E}(Y^2) < \infty$.

Throughout this proof, we use the Cauchy–Schwarz inequality for expectations: $|\mathrm{E}(AB)| \le \sqrt{\mathrm{E}(A^2)} \cdot \sqrt{\mathrm{E}(B^2)}$, which follows from the same technique as Question 2(i).

- **$|\mathbf{E}X| < \infty$ and $|\mathbf{E}Y| < \infty$:**

  Proof. Apply Cauchy–Schwarz with $B = 1$:

  $$|\mathrm{E}X| = |\mathrm{E}(X \cdot 1)| \le \sqrt{\mathrm{E}(X^2)} \cdot \sqrt{\mathrm{E}(1^2)} = \sqrt{\mathrm{E}(X^2)} < \infty.$$

  Analogously for Y. □

  Alternative (via Jensen): Since $x \mapsto x^2$ is convex, Jensen's inequality gives $(\mathrm{E}X)^2 \le \mathrm{E}(X^2) < \infty$, so $|\mathrm{E}X| \le \sqrt{\mathrm{E}(X^2)} < \infty$.

- **$|\mathbf{E}(XY)| < \infty$:**

  Proof. By Cauchy–Schwarz:

  $$|\mathrm{E}(XY)| \le \sqrt{\mathrm{E}(X^2)} \cdot \sqrt{\mathrm{E}(Y^2)} < \infty. \square$$

- **$|\mathbf{Cov}(X, Y)| < \infty$:**

  Proof. Since $\mathrm{Cov}(X, Y) = \mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)$, the triangle inequality gives:

  $$|\mathrm{Cov}(X, Y)| \le |\mathrm{E}(XY)| + |\mathrm{E}X| \cdot |\mathrm{E}Y| < \infty,$$

  where each term is finite by the results above. □

Discussion. The key insight is that L_2 (finite second moments) is the natural space for covariance analysis. Everything we need—means, cross-moments, covariances—is guaranteed to exist once we have finite second moments. This is why L_2 is the workhorse space in econometrics.

Ex 4

Lemma: Law of iterated expectations (LIE)

$$E(Y) = E\big(E(Y|X)\big)$$

$$\text{Cov}(X,Y) := E(XY) - E(X)\,E(Y)$$

now, using LIE:

$$= E\big(E(XY|X)\big) - E(X)\,E\big(E(Y|X)\big)$$

$$= E\big(X \cdot E(Y|X)\big) - E(X) \cdot E\big(E(Y|X)\big)$$

$$= \text{Cov}(X, Z)$$

where $Z := E(Y|X)$

Ex 5

Notice $X \sim$ Bernoulli with $\pi := P(X=1)$

recall: $\text{Var}(X) = \pi(1-\pi)$

using Ex 4:

$\text{Cov}(X,Y)$

either 0 or 1

$$= E\big(X \cdot E(Y|X)\big) - E(X) \cdot E\big(E(Y|X)\big)$$

$$= \pi \cdot E(Y|X=1) - \pi \cdot \big[(1-\pi)E(Y|X=0) + \pi\,E(Y|X=1)\big]$$

$$= \pi(1-\pi)\big[E(Y|X=1) - E(Y|X=0)\big]$$

4. **Cov**$(X,Y) =$ **Cov**$(X, \mathbf{E}(Y|X))$.

   Proof. Expand the right-hand side using the definition of covariance:

   $$\mathrm{Cov}(X, \mathrm{E}(Y|X)) = \mathrm{E}\big[X \cdot \mathrm{E}(Y|X)\big] - \mathrm{E}(X) \cdot \mathrm{E}\big[\mathrm{E}(Y|X)\big].$$

   Second term. By the law of iterated expectations (LIE): $\mathrm{E}[\mathrm{E}(Y|X)] = \mathrm{E}(Y)$.

   First term. Since X is measurable with respect to $\sigma(X)$, the "taking out what is known" property of conditional expectation gives: $X \cdot \mathrm{E}(Y|X) = \mathrm{E}(XY|X)$. Applying LIE:

   $$\mathrm{E}\big[X \cdot \mathrm{E}(Y|X)\big] = \mathrm{E}\big[\mathrm{E}(XY|X)\big] = \mathrm{E}(XY).$$

   Combining:

   $$\mathrm{Cov}(X, \mathrm{E}(Y|X)) = \mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y) = \mathrm{Cov}(X,Y) \quad \square$$

   Discussion. This result says that the covariance between X and Y is entirely captured by the covariance between X and the conditional expectation $\mathrm{E}(Y|X)$. In other words, the "noise" $Y - \mathrm{E}(Y|X)$ is uncorrelated with X. This foreshadows the projection idea: the conditional expectation extracts all the information in Y that is linearly (and nonlinearly) related to X.

5. **Binary regressor:** $\frac{\mathbf{Cov}(X,Y)}{\mathbf{Var}X} = \mathbf{E}(Y|X=1) - \mathbf{E}(Y|X=0)$.

   Proof. Let $p := P(X=1)$, so $P(X=0) = 1-p$.

   Moments of X: Since $X \in \{0,1\}$, we have $X^2 = X$, so:

   $$\mathrm{E}X = p, \qquad \mathrm{E}(X^2) = p, \qquad \mathrm{Var}X = p - p^2 = p(1-p).$$

   Cross-moment $E(XY)$: By the law of total expectation:

   $$\begin{aligned} \mathrm{E}(XY) &= \mathrm{E}(XY|X=1)\,p + \mathrm{E}(XY|X=0)\,(1-p) \\ &= 1 \cdot \mathrm{E}(Y|X=1)\,p + 0 \cdot \mathrm{E}(Y|X=0)\,(1-p) \\ &= p\,\mathrm{E}(Y|X=1). \end{aligned}$$

   Mean of Y:

   $$\mathrm{E}Y = \mathrm{E}(Y|X=1)\,p + \mathrm{E}(Y|X=0)\,(1-p).$$

   Covariance:

   $$\begin{aligned} \mathrm{Cov}(X,Y) = \mathrm{E}(XY) - \mathrm{E}X \cdot \mathrm{E}Y & \\ &= p\,\mathrm{E}(Y|X=1) - p\big[p\,\mathrm{E}(Y|X=1) + (1-p)\,\mathrm{E}(Y|X=0)\big] \\ &= p(1-p)\,\mathrm{E}(Y|X=1) - p(1-p)\,\mathrm{E}(Y|X=0) \\ &= p(1-p)\big[\mathrm{E}(Y|X=1) - \mathrm{E}(Y|X=0)\big]. \end{aligned}$$

   Dividing by $\mathrm{Var}X = p(1-p)$:

   $$\frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}X} = \mathrm{E}(Y|X=1) - \mathrm{E}(Y|X=0) \quad \square$$

Discussion. This is a fundamental result for causal inference. The population regression coefficient of Y on a binary X equals the difference in conditional means. When X is a randomly assigned treatment indicator, $\mathrm{E}(Y|X=1) - \mathrm{E}(Y|X=0)$ is the average treatment effect. So OLS with a binary regressor recovers the difference-in-means estimator.

6. **E(XY) is an inner product on L_2.**

We need to verify the three axioms of an inner product: (1) linearity, (2) symmetry, and (3) positive definiteness.

Recall: elements of L_2 are equivalence classes of random variables, where $X \sim Y$ iff $X = Y$ almost surely.

Proof. *(1) Linearity in the first argument.* For $a, b \in \mathbb{R}$ and $X_1, X_2, Y \in L_2$:

$$\mathrm{E}\big[(aX_1 + bX_2)\,Y\big] = a\,\mathrm{E}(X_1Y) + b\,\mathrm{E}(X_2Y),$$

$E(XY) = E(YX)$

$E((X+Y)Z) = E(XZ) + E(YZ)$

$E(\lambda X Z) = \lambda E(XZ)$

by linearity of the expectation operator.

(2) Symmetry. For $X, Y \in L_2$:

$$\mathrm{E}(XY) = \mathrm{E}(YX),$$

since multiplication of real numbers is commutative.

(3) Positive definiteness. For $X \in L_2$:

$E(XX) = E(X^2) \geq 0$

$E(XX) = 0 \iff X = 0$

$$\mathrm{E}(X^2) \geq 0,$$

since $X^2 \geq 0$ pointwise. Moreover, $\mathrm{E}(X^2) = 0$ if and only if $X = 0$ almost surely, which means X is the zero element in L_2. $\square$

We should also verify that $\mathrm{E}(XY)$ is *well-defined* (finite) for all $X, Y \in L_2$: by Cauchy–Schwarz, $|\mathrm{E}(XY)| \leq \sqrt{\mathrm{E}(X^2)\,\mathrm{E}(Y^2)} < \infty$.

7. **Projection of Y on sp(X_2, X_3) via Gram–Schmidt.**

We are *not* including a constant. The inner product is $\langle A, B\rangle = \mathrm{E}(AB)$ and $\|A\| = \sqrt{\mathrm{E}(A^2)}$.

Step 1: Normalize X_2. Set $V_1 = X_2$.

$$U_1 = \frac{X_2}{\|X_2\|} = \frac{X_2}{\sqrt{\mathrm{E}(X_2^2)}}.$$

Step 2: Orthogonalize X_3 against U_1 and normalize. Set $V_2 = X_3$.

$$\langle X_3, U_1\rangle = \mathrm{E}\left(X_3 \cdot \frac{X_2}{\sqrt{\mathrm{E}(X_2^2)}}\right) = \frac{\mathrm{E}(X_2X_3)}{\sqrt{\mathrm{E}(X_2^2)}}.$$

The orthogonalized (but not yet normalized) version:

$$\ddot{X}_3 := X_3 - \langle X_3, U_1\rangle\, U_1 = X_3 - \frac{\mathrm{E}(X_2X_3)}{\mathrm{E}(X_2^2)}\, X_2.$$

$$P_{sp(X_2, X_3)}\, y = \sum_{i=2}^{3} \langle \tilde{X}_i, y \rangle \tilde{X}_i = \sum_{i=2}^{3} E(\tilde{X}_i \cdot y)\, \tilde{X}_i$$

$$= \sum_{i=2}^{3} \frac{E(\ddot{X}_i\, y)\, \ddot{X}_i}{\|\ddot{X}_i\|^2}$$

<u>Gram - Schmidt</u>

<u>notation:</u>

$$m_{ij} := E(X_i X_j)$$

$$m_{iy} := E(X_i\, y)$$

$$\ddot{X}_2 = X_2$$

$$\tilde{X}_2 = X_2 / \|X_2\|$$

$$\ddot{X}_3 = X_3 - E(\tilde{X}_2 X_3)\, \tilde{X}_2 = X_3 - \frac{m_{23}}{m_{22}} X_2$$

$$\|\ddot{X}_3\|^2 = m_{33} - \frac{2 m_{23}^2}{m_{22}} + \frac{m_{23}^2}{m_{22}} = \frac{m_{22} m_{33} - m_{23}^2}{m_{22}}$$

<u>Therefore:</u>

$$P_{sp(X_2, X_3)}\, y = \frac{E(X_2\, y)\, X_2}{m_{22}} + \frac{E(\ddot{X}_3\, y)\, \ddot{X}_3}{\|\ddot{X}_3\|^2}$$

$$= \frac{m_{2y}}{m_{22}} X_2 + \frac{m_{22}}{m_{22} m_{33} - m_{23}^2} \cdot \frac{m_{22} m_{3y} - m_{23} m_{2y}}{m_{22}} \left(X_3 - \frac{m_{23}}{m_{22}} X_2 \right)$$

$$= \frac{m_{2y}}{m_{22}} X_2 - \frac{m_{22} m_{3y} - m_{23} m_{2y}}{m_{22} m_{33} - m_{23}^2} \cdot \frac{m_{23}}{m_{22}} X_2$$

$$+ \frac{m_{22} m_{3Y} - m_{23} m_{2Y}}{m_{22} m_{33} - m_{23}^2} X_3$$

$$= \frac{(\cancel{m_{22}} m_{33} - \cancel{m_{23}^2}) m_{2Y} - (\cancel{m_{22}} m_{3Y} - \cancel{m_{23} m_{2Y}}) m_{23}}{\cancel{m_{22}} (m_{22} m_{33} - m_{23}^2)} X_2$$

$$+ \frac{m_{22} m_{3Y} - m_{23} m_{2Y}}{m_{22} m_{33} - m_{23}^2} X_3$$

$$= \underbrace{\frac{m_{33} m_{2Y} - m_{23} m_{3Y}}{m_{22} m_{33} - m_{23}^2}}_{\beta_2^*} X_2 + \underbrace{\frac{m_{22} m_{3Y} - m_{23} m_{2Y}}{m_{22} m_{33} - m_{23}^2}}_{\beta_3^*} X_3$$

Contrast this with the case in which $X_1 = \text{const}$

$$\beta_2^* := \frac{\sigma_{2Y}\sigma_3^2 - \sigma_{3Y}\sigma_{23}}{\sigma_2^2\sigma_3^2 - \sigma_{23}^2}$$

$$\beta_3^* := \frac{\sigma_{3Y}\sigma_2^2 - \sigma_{2Y}\sigma_{23}}{\sigma_2^2\sigma_3^2 - \sigma_{23}^2}$$

$$\beta_1^* := EY - \beta_2^* EX_2 - \beta_3^* EX_3$$

Normalize: $U_2 = \ddot{X}_3 / \left\| \ddot{X}_3 \right\|$.

Note: $\ddot{X}_3$ is the "residual" of X_3 after projecting out X_2. By construction, $\langle \ddot{X}_3, U_1 \rangle = 0$, so $\{U_1, U_2\}$ is orthonormal.

Step 3: Project Y. Since sp$(U_1, U_2) =$ sp(X_2, X_3):

$$\hat{Y} = \mathbb{P}_{\text{sp}(X_2,X_3)} Y = \langle Y, U_1 \rangle U_1 + \langle Y, U_2 \rangle U_2.$$

Expanding each term:

$$\langle Y, U_1 \rangle U_1 = \frac{\mathrm{E}(Y X_2)}{\mathrm{E}(X_2^2)} X_2,$$

$$\langle Y, U_2 \rangle U_2 = \frac{\mathrm{E}(Y \ddot{X}_3)}{\mathrm{E}(\ddot{X}_3^2)} \ddot{X}_3 = \frac{\mathrm{E}(Y \ddot{X}_3)}{\mathrm{E}(\ddot{X}_3^2)} \left(X_3 - \frac{\mathrm{E}(X_2 X_3)}{\mathrm{E}(X_2^2)} X_2 \right).$$

Therefore, $\hat{Y} = \beta_2^* X_2 + \beta_3^* X_3$, where:

$$\beta_3^* = \frac{\mathrm{E}(Y \ddot{X}_3)}{\mathrm{E}(\ddot{X}_3^2)}, \qquad \text{with } \ddot{X}_3 = X_3 - \frac{\mathrm{E}(X_2 X_3)}{\mathrm{E}(X_2^2)} X_2,$$

$$\beta_2^* = \frac{\mathrm{E}(Y X_2)}{\mathrm{E}(X_2^2)} - \beta_3^* \cdot \frac{\mathrm{E}(X_2 X_3)}{\mathrm{E}(X_2^2)}.$$

Discussion. The coefficient β_3^* has a clean interpretation: it is the coefficient from projecting Y onto $\ddot{X}_3$, the part of X_3 that is orthogonal to X_2. This is exactly the Frisch–Waugh–Lovell idea: to get the coefficient on X_3, first "partial out" X_2 from X_3, then project Y onto the residual.

Verification via normal equations. Alternatively, the projection satisfies the orthogonality conditions $\mathrm{E}(X_k(\hat{Y} - Y)) = 0$ for $k = 2, 3$. Stacking $X = (X_2, X_3)'$, this gives $\mathrm{E}(XX')\beta^* = \mathrm{E}(XY)$, i.e.

$$\begin{pmatrix} \mathrm{E}(X_2^2) & \mathrm{E}(X_2 X_3) \\ \mathrm{E}(X_2 X_3) & \mathrm{E}(X_3^2) \end{pmatrix} \begin{pmatrix} \beta_2^* \\ \beta_3^* \end{pmatrix} = \begin{pmatrix} \mathrm{E}(X_2 Y) \\ \mathrm{E}(X_3 Y) \end{pmatrix}.$$

This is equivalent to the Gram–Schmidt answer (one can verify by substituting back).

8. **Deriving $\tilde{\beta}$ via calculus.**

   We want to solve:

$$\tilde{\beta} := \operatorname*{argmin}_{b \in \mathbb{R}^K} \mathrm{E}\left((Y - X'b)^2 \right),$$

   where $X = (X_1, \ldots, X_K)'$ is $K \times 1$ and $b \in \mathbb{R}^K$.

   Expand the objective:

$$\begin{aligned} Q(b) := \mathrm{E}\left((Y - X'b)^2 \right) &= \mathrm{E}\left(Y^2 - 2Yb'X + b'XX'b \right) \\ &= \mathrm{E}(Y^2) - 2b'\mathrm{E}(XY) + b'\mathrm{E}(XX')b. \end{aligned}$$

First-order condition (FOC):

$$\frac{\partial Q}{\partial b} = -2\mathrm{E}(XY) + 2\mathrm{E}(XX')\, b \overset{!}{=} 0.$$

Solving:

$$\mathrm{E}(XX')\,\tilde{\beta} = \mathrm{E}(XY) \qquad \Longrightarrow \qquad \tilde{\beta} = [\mathrm{E}(XX')]^{-1}\,\mathrm{E}(XY),$$

assuming $\mathrm{E}(XX')$ is invertible (i.e., no perfect collinearity among $X_1, \ldots, X_K$).

Second-order condition (SOC):

$$\frac{\partial^2 Q}{\partial b\, \partial b'} = 2\,\mathrm{E}(XX'),$$

which is positive semi-definite (positive definite under the invertibility assumption), confirming that $\tilde{\beta}$ is a minimizer.

Discussion. The FOC can be rewritten as:

$$\mathrm{E}\Big(X(Y - X'\tilde{\beta})\Big) = 0.$$

This is exactly the orthogonality condition from projection theory: the residual $Y - X'\tilde{\beta}$ is orthogonal (in the L_2 inner product) to every element of $\mathrm{sp}(X_1, \ldots, X_K)$. So the calculus approach and the geometric projection approach yield the same answer. This is the population analogue of the familiar OLS normal equations $\boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{X}\hat{\beta}) = 0$.